Partial ports and crossing fluctuations in the three-dimensional Lorentz mirror model

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Abstract

Deterministic trajectories in random mirror environments produce fluctuations between environments. We study their dependence on which boundary ports are counted. Pure retroreflection on part of a face admits an exact environment-by-environment reduction to selected connections in a fully open reference system. Applying established multiterminal diffusive counting theory gives a conditional variance-to-mean ratio $1 - J/M$, determined by a Laplace problem. For m stripes occupying fraction p of a face, Fourier evaluation gives $1 - p/3 - [\zeta(3) - \Re\,\mathrm{Li}_3(e^{2\pi i p})]/(4\pi^3 mp)$, up to exponentially small terms. Two mirror rules and sizes up to $L = 256$ support the half-open predictions within the reported statistical errors and extrapolation spread. The two prescribed extrapolations differ by approximately 0.3–0.7%. Earlier tests reject 2/3 for the retroreflecting count, while its reconstructed full-face count is consistent with that benchmark. Exploratory quarter-open layouts lie about 0.5% high; independent third-cumulant estimates lie on opposite sides of the saddle value. These discrepancies remain unresolved. Our contribution concerns this specific reduction, three-dimensional geometry and numerical tests; partial-boundary counting in loop models and the general multiterminal framework have precedents.

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1 Introduction

The Lorentz mirror model separates deterministic propagation from random environment selection. Paired ports prescribe a unique outgoing direction for each incoming ray. Statistics concern different environments, despite the absence of dynamical randomness in any one of them. Deterministic lattice transport and macroscopic diffusion have substantial precedents [1, 2]. Lefevere and Tasaki discuss a 2/3 mean–variance relation and higher crossing cumulants [3]. Loop models with crossings provide a related counting-field description [4].

The theoretical setting includes Goldstone phases of intersecting loops [5], oriented three-dimensional loops [6, 7], and unoriented loops with a real-projective description [8]. These results motivate a smooth candidate theory; they do not prove its complete applicability to both six-valent rules used here.

Diffusive-conductor counting statistics are established [9, 10, 11, 12, 13]. Multiterminal noise and full counting statistics were developed by Sukhorukov–Loss and Nazarov–Bagrets [14, 15, 16]. Related structures occur in stochastic diffusive systems [17, 18, 19]. Akkermans and coauthors extend two-reservoir counting to arbitrary higher-dimensional domains within macroscopic fluctuation theory, with macroscopic contacts and a stationary saddle [20]. Here the random object is a quenched environment rather than a time-integrated stochastic current.

Partial-boundary counting also has a direct loop-model precedent. Morin-Duchesne and Jacobsen weight connections between a chosen boundary segment and its complement in a two-dimensional dense-loop model [21]. That geometry and count differ from the present left-face-to-right-subset observable. Gu, Schaeffer and Hao study fixed-aperture routing and suppression of nearby transmitting channels in a three-dimensional oriented K-lattice [22]. Boundary spatial correlations are therefore not a new general observation of this work.

Within the literature we searched, the contribution here is the specific pure-retroreflection reduction, its three-dimensional stripe evaluation and quantitative tests. We assert neither a new general counting framework nor a new universality class. Classical representations of quantum localization [23, 24, 25] do not in general equate the fluctuations of the two problems, as emphasized by Cardy [26].

2 Model and observables

Each cubic-lattice vertex independently receives a uniform perfect matching of its six ports. The standard rule permits all 15 matchings; the orthogonal rule permits the eight without an opposite-port pair. Transport is along the open x direction; transverse directions are periodic.

Let L_t denote the full left terminal, distinct from side length L. The right face is partitioned into counted ports A and complement W. Write $C = n_{L_\mathrm{t}A}$, and let N count all left-to-right paths with the entire face open. Cumulants refer to independent environments:

$$v = \kappa_2(C)/\kappa_1(C), \qquad h = \kappa_3(C)/\kappa_1(C).$$

An m-stripe mask has transverse period $1/m$ and total area fraction p. All measured sizes resolve the discrete masks exactly. For every terminal,

$$2n_{\alpha\alpha} + \sum_{\beta\neq\alpha} n_{\alpha\beta} + f_\alpha = A_\alpha. \tag{1}$$

Here f_α counts a ray returning to its own starting port after retroreflection. This consumes one terminal port, whereas a paired return consumes two. Retroreflecting counts can consequently be odd.

3 Exact pure-retroreflection reduction

Open W while keeping every bulk matching fixed. A reference-graph strand is closed or joins two boundary ports. Closing a strand end by pure retroreflection reverses the same path. Left-to-A strands survive; left-to-W strands return to their own starting ports. Therefore, in each environment,

$$C_\mathrm{retro} = n^{\mathrm{open}}_{L_\mathrm{t}A}, \qquad f_{L_\mathrm{t}} = n^{\mathrm{open}}_{L_\mathrm{t}W}, \qquad N = C_\mathrm{retro} + f_{L_\mathrm{t}}. \tag{2}$$

No continuum approximation is used. This concerns pure retracing, not walls pairing distinct ports. Environment weights remain fixed; reweighting loops after changing a boundary would require a separate argument.

4 Conditional diffusive description

4.1 Finite source and inherited counting theory

The local link-spin weight

$$W_r = |P_r|^{-1} \sum_{P\in P_r} \prod_{(a,b)\in P} S_a \cdot S_b, \qquad \int d\mu(S) S^a S^b = \delta_{ab}$$

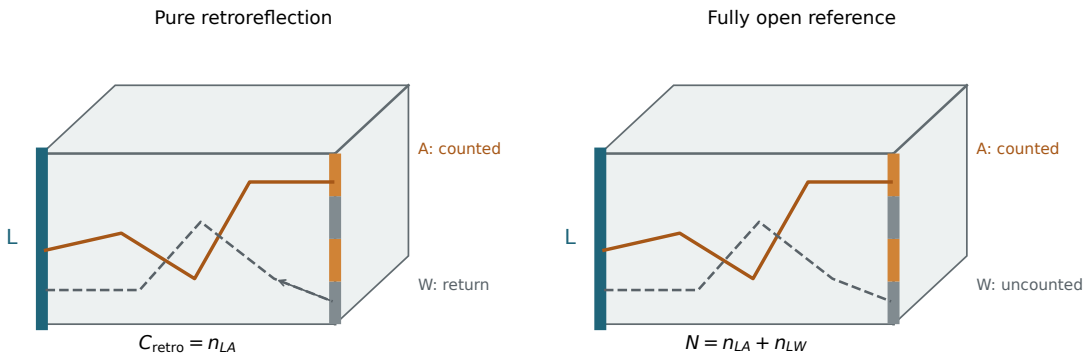


Figure 1: Opening the retroreflecting complement W converts retraced returns to left-to-W connections. Paths counted from the full left face to A are unchanged. The bulk environment is fixed.

contracts to a factor n per closed loop and a boundary scalar product per open path. Evaluating the finite polynomial at $n = 1$ recovers the mirror ensemble. Using bilinear complex products, take

$$u_L = (1, a, ia), \quad u_A = (1, b, -ib), \quad u_W = (1, 0, 0), \quad z = 1 + 2ab.$$

Only $u_L \cdot u_A = z$ differs from one, yielding $\mathbb{E}z^C$. Finite enumeration and independent tensor contraction check this identity. They do not establish the infinite-volume field theory.

Assume an ordered smooth sector, its leading isotropic gradient action and the prescribed boundary data. Near zero source its stationary action matches diffusive conductor counting theory [16] under

$$q = (1 + z)/2, \quad F_C(s) = F_e\left(\log \frac{1 + e^s}{2}\right), \quad v_C = (1 + v_e)/2, \quad h_C = (3v_e + h_e)/4, \tag{3}$$

with conductor mean twice the mirror mean. This is a formal generating-function relation within the saddle. It does not identify microscopic probability measures, fluctuation determinants, topological sectors or physical trajectories.

4.2 Boundary functional

Let ϕ_L, ϕ_A be harmonic characteristic potentials, one on their own terminal and zero on the other open terminals, in the fully open reference. Define

$$M = -\int_\Omega \nabla\phi_L \cdot \nabla\phi_A, \qquad J = \int_\Omega |\nabla(\phi_L\phi_A)|^2. \tag{4}$$

The characteristic-potential noise integral [14] and Eq. (3) give

$$\log \mathbb{E}e^{sC} \simeq KM(e^s - 1) - \frac{KJ}{2}(e^s - 1)^2 + \cdots, \qquad v = 1 - J/M. \tag{5}$$

The stiffness K cancels. This general structure is inherited from prior theory; the Supplement gives the boundary-matrix comparison.

For separated right terminals with potentials ϕ, ψ, define M, J in the same way. The mixed-twist coefficient predicts $\mathrm{Cov}(X, Y)/\mathbb{E}n_{R_1R_2} = 2J/M$, where $X = n_{LR_1} + n_{R_1R_2}$, $Y = n_{LR_2} + n_{R_1R_2}$. Neumann conditions on a side-pairing wall are an additional microscopic assumption. These tests are distinct from the exact retroreflection reduction.

4.3 Stripe closed expression

For a unit cube $\phi_L = 1 - x$, $M = p$, and integration by parts gives $J = \int \phi_A^2$. For mask Fourier coefficients f_n,

$$J = \frac{p^2}{3} + \sum_{n \neq 0} |f_n|^2 I(2\pi|n|), \qquad I(k) = \int_0^1 \frac{\sinh^2(kx)}{\sinh^2 k}\, dx = \frac{\coth k}{2k} - \frac{1}{2\sinh^2 k}. \tag{6}$$

For m stripes, nonzero frequencies are mj, with squared coefficients $\sin^2(\pi jp)/(\pi j)^2$. Separating $I(k) = 1/(2k) + O(e^{-2k})$ gives, for fixed $0 < p \leq 1$,

$$v(p, m) = 1 - \frac{p}{3} - \frac{\zeta(3) - \Re\,\mathrm{Li}_3(e^{2\pi ip})}{4\pi^3 mp} + O(e^{-4\pi m}). \tag{7}$$

At $p = 1/2$, the leading expression is $5/6 - 7\zeta(3)/(8\pi^3 m)$. Numerical targets retain the exponential correction: about 2.60×10^{-6} for $m = 1$, and below 10^{-11} for $m = 2$. The full face gives $2/3$ exactly; the $m \to \infty$ limit at fixed p is $1 - p/3$. The latter matches independent binomial thinning of a count with ratio $2/3$, but does not establish thinning for a fixed mask.

5 Numerical design

The initial study used $L = 16, 32, 64, 96$, both rules and several domain and wall geometries. All 30 planned tests and sample allocations are in the Supplement. The subsequent layout data set contains 2,522,250 environments at $L = 16, 32, 64, 128$, with paired primary contrast $v_4 - v_1$. The fresh absolute-ratio data set contains 1,146,500 environments at $L = 32, 64, 128, 256$. Old and pilot samples do not enter that primary analysis.

Rules, sample counts and seeds were fixed and hashed before the respective confirmation samples. These were *local* preregistrations, without a third-party timestamp or registry. Public hashes verify file contents, not independently when the rules were fixed.

Both $a + b/L$ and $a + b/L + c/L^2$ are retained for every analysis. These are prescribed fitting assumptions, not a derived expansion with bounded remainder. Joint fits share only the intercept; each rule has its own correction coefficients. Bootstrap resampling uses independent environments, preserves paired correlations and keeps the original fixed bootstrap seeds. No sample or size is removed because of its result.

For a half-open mask, let Q_m average $(C_m - N/2)^2$ over all transverse translations and both orientations within an environment. Symmetry gives

$$\mathrm{Var}(C_m) = \tfrac{1}{4}\,\mathrm{Var}(N) + \mathbb{E}Q_m, \qquad \widehat{v}_m = (\tfrac{1}{2}s_N^2 + 2\overline{Q}_m)/\overline{N}. \tag{8}$$

Translated masks are not independent observations. Direct cyclic-window sums checked the Fourier implementation on every saved environment. The absolute-ratio analysis uses six Holm tests at level 0.05, fit probabilities at least 0.01, rule differences within three combined standard errors, and joint errors at most 0.003. Consistency under these checks is not an equivalence proof.

6 Results

6.1 Half-open stripes

All three targets lie between the joint linear and quadratic estimates. The linear deviations are about $0.3, 0.6, 0.9$ standard errors upward; the quadratic deviations are about $1.0, 1.5, 1.9$ downward. All prescribed checks pass. The differences between forms are approximately $0.0027, 0.0042, 0.0056$, exceeding the linear statistical errors. This spread describes extrapolation sensitivity; it is not a confidence interval or a complete systematic-error bound.

Table 1: Half-open stripe ratios. Errors are bootstrap standard errors; both prescribed extrapolations use all four sizes.

m	Theory	Linear	Quadratic
1	0.799413778	0.799752 ± 0.001116	0.797022 ± 0.002399
2	0.816372255	0.817086 ± 0.001130	0.812874 ± 0.002408
4	0.824852794	0.825861 ± 0.001150	0.820215 ± 0.002456

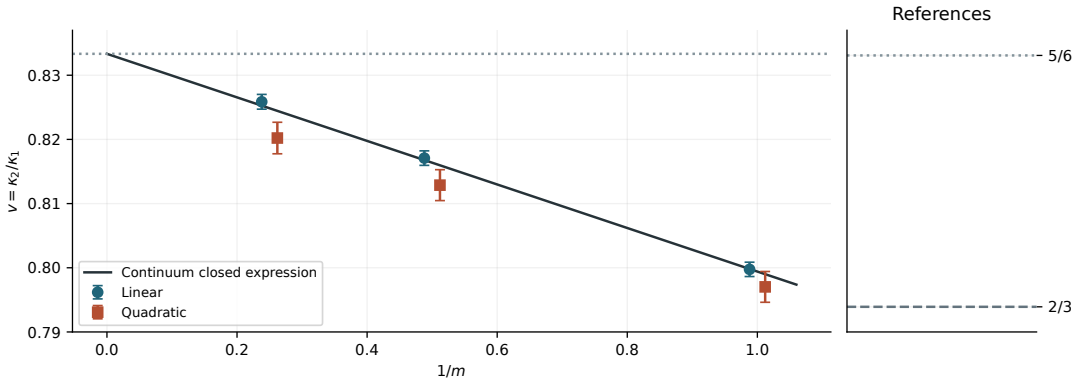


Figure 2: Half-open ratios versus inverse stripe number, with one-standard-error bootstrap bars. The line omits the exponential term; tabulated targets retain it. Horizontal offsets distinguish the two extrapolations. The right panel uses a wider vertical range to show both limiting reference values.

The independent layout data reject zero difference between four and one stripes under the original rules. Joint estimates are $0.02671905 \pm 0.00210647$ and $0.02514860 \pm 0.00532498$, versus the predicted 0.0254390161. This supports shape dependence without establishing every absolute coefficient or arbitrary boundary geometry.

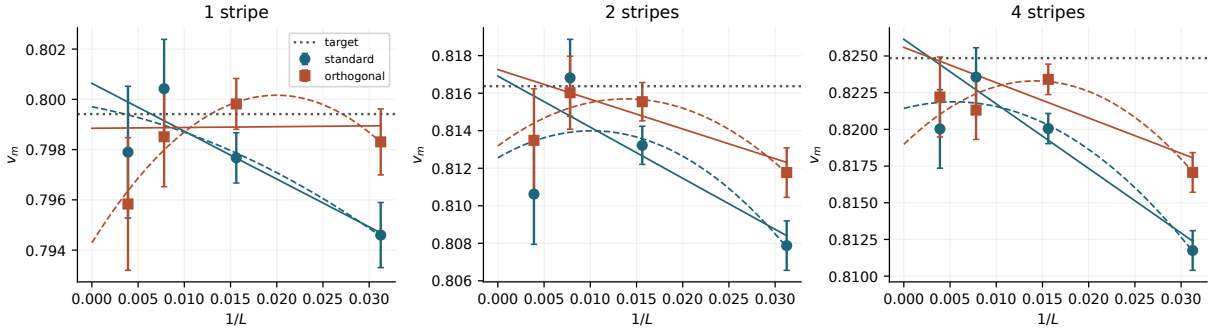


Figure 3: All fresh half-open data, both rules and both prescribed separate-rule fits (solid: linear; dashed: quadratic). No small-size points are discarded.

6.2 Other boundaries

The fully open cube is consistent with 2/3; separated-terminal covariance is consistent with its conditional target for both wall wirings. Five of the twelve related covariance-identity tests remain undecided. The constricted-domain second-cumulant fit and one wall's third-cumulant fit fail their adequacy checks. Fourth-cumulant tests remain inconclusive.

The original retroreflecting count robustly rejects 2/3. Its later exploratory comparison to the two-stripe prediction gives deviations between approximately -0.70 and 2.65 standard errors.

Table 2: Original boundary tests, with one-standard-error uncertainties. Cube and retroreflection were tested against 2/3. The covariance target for each gapped wall is 0.5138587 with a deterministic allowance 0.0018720, distinct from a statistical error.

Geometry / observable	Rule	Linear	Quadratic
Cube	Standard	0.670186 ± 0.002851	0.666812 ± 0.007241
Cube	Orthogonal	0.669290 ± 0.003060	0.669803 ± 0.006917
Gap, wall A	Standard	0.517004 ± 0.001513	0.514276 ± 0.003236
Gap, wall A	Orthogonal	0.514532 ± 0.001441	0.514961 ± 0.003056
Gap, wall B	Standard	0.517263 ± 0.001422	0.517552 ± 0.003278
Gap, wall B	Orthogonal	0.517928 ± 0.001388	0.517246 ± 0.002997
Retroreflection	Standard	0.822143 ± 0.002181	0.822773 ± 0.004819
Retroreflection	Orthogonal	0.818848 ± 0.002116	0.813176 ± 0.004597

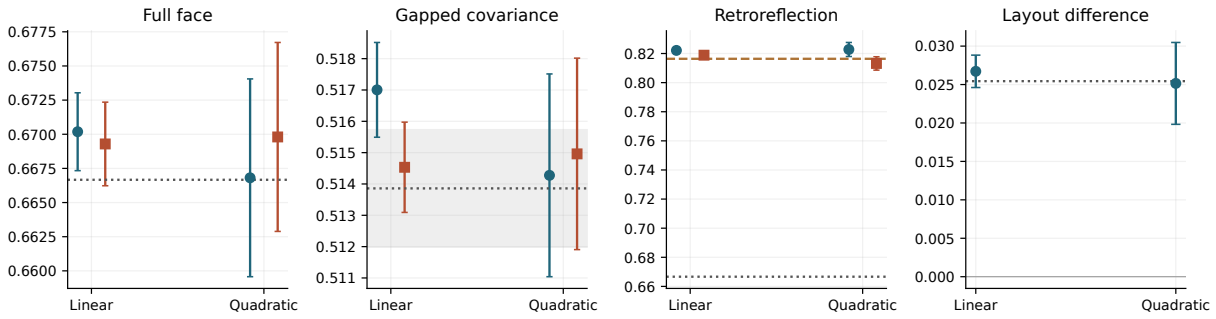


Figure 4: Selected original boundary tests and the independent layout contrast. Bars are one standard error. The covariance band is the separate deterministic theory allowance. Teal circles denote the standard rule and rust squares the orthogonal rule; the layout contrast uses joint fits. Retroreflection is shown against both its original 2/3 null and the later partial-port prediction (dashed).

Reconstructing $N = C + f_L$ from the same data gives full-face intercepts between 0.66650 and 0.67085. These are retrospective comparisons, not new independent samples.

6.3 Residual discrepancies

The older standard-rule layout data have a shared linear-intercept excess 0.007056 ± 0.001616. Pooling both batches and all sizes leaves a one-stripe linear excess of about 3.4–3.5 standard errors under two pooling conventions, while quadratic estimates remain near theory. Small-size curvature is a possible retrospective explanation, not an established microscopic correction.

Table 3: Exploratory quarter-open results. Joint fits are descriptive; neither batch nor rule is discarded.

Observable	Target	Linear	Quadratic
Quarter strip	0.874266572	0.878925 ± 0.001518	0.878854 ± 0.002585
Square	0.872115450	0.876012 ± 0.001484	0.876146 ± 0.002530
Square minus strip	-0.002151122	-0.002922 ± 0.000709	-0.002746 ± 0.001201

The quarter-open layouts use 196,000 diagnostic environments including $L = 16$, and are exploratory. Their difference is consistent with the geometric prediction, but both absolute values are about 0.5% high. The strip's linear estimate is about 3.07 standard errors above theory. Neither form, rule nor layout is omitted.

The one-stripe third-cumulant target is 0.4994424165 within the saddle. Fresh half-open data give 0.4735764 ± 0.0101997 linearly and 0.4797957 ± 0.0305811 quadratically; independent diagnostic data give 0.5150117 ± 0.0176500 and 0.5059868 ± 0.0380576. The linear estimates lie on opposite sides and differ by about two combined standard errors; the size ranges differ. Within each batch, all three masks share the environments and the full-face third-cumulant term, so they are not three independent pieces of evidence. These auxiliary results supply no new confirmation decision.

7 Limitations and open questions

The graph reduction is exact independently of continuum theory. The stripe predictions additionally assume a smooth diffusive phase and appropriate boundary data. Existing transport and loop results, and the present tests, do not prove those assumptions for the two microscopic rules. The formal counting dictionary does not identify quantum time statistics with quenched sample statistics.

There is no controlled remainder bounding all omitted bulk, surface, contact-line and corner terms. Exploratory matching included microscopic fourth- and sixth-order sources and specified one- and two-loop contributions. It did not produce a complete physical error budget or explain the older third-cumulant downward discrepancy. The Supplement records the scope and failures of that attempt. Those corrections are not applied to the primary results or their error bars.

Quarter-open absolute ratios need independent confirmation before supporting a broader claim about $p \neq 1/2$. Their deviations do not establish failure of the smooth description or justify neglecting finite sizes. Higher cumulants, unresolved covariance identities and finite-precision controls remain open. So do a rigorous long-distance limit, defect control, and interchange of replica continuation, source differentiation and volume limits. This is an exact boundary reduction with conditional tested predictions, not a completed microscopic field theory.

Data, code and contributions

The accompanying release candidate includes compact observations for every environment in both stripe data sets, original seeds and local registration files, summaries and reproduction scripts. Full original blocks and historical failed runs remain separately preserved. Compact records support the primary analyses, not every auxiliary higher-order response.

JJIN HANZHE provided directional guidance. Opus 5.5 proposed the initial plan; GPT-6 Astra (max) performed later design, simulations, analysis, derivations and writing. Checks use separate implementations, exact identities and independent numerical calculations, not agreement between AI systems. There has been no external human expert review. Code is licensed MIT; manuscript and data are licensed CC BY 4.0.

A Fourier identity and omitted term

The harmonic profile $\sinh(kx)/\sinh k$ directly integrates to Eq. (6). The zero mode px contributes $p^2/3$. With $r = e^{-2k}$,

$$I(k) - \frac{1}{2k} = \frac{r}{k(1-r)} - \frac{2r}{(1-r)^2}, \qquad \sum_{j\geq 1} \frac{\sin^2(\pi jp)}{j^3} = \frac{\zeta(3) - \Re\,\mathrm{Li}_3(e^{2\pi ip})}{2}.$$

For rational p, an independent Hurwitz-zeta residue-class sum and numerical kernel integration reproduce the targets. The square window requires a two-dimensional sum with an explicit positive tail bound. Neither numerical bound controls microscopic finite-size effects.

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